

VSR-Net: Vessel-Like Structure Rehabilitation Network With Graph Clustering

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Abstract—The morphologies of vessel-like structures, such as blood vessels and nerve fibres, play significant roles in disease diagnosis, e.g., Parkinson’s disease. Although deep network-based refinement segmentation and topology-preserving segmentation methods recently have achieved promising results in segmenting vessel-like structures, they still face two challenges: 1) existing methods often have limitations in rehabilitating subsection ruptures in segmented vessel-like structures; 2) they are typically overconfident in predicted segmentation results. To tackle these two challenges, this paper attempts to leverage the potential of spatial interconnection relationships among subsection ruptures from the structure rehabilitation perspective. Based on this perspective, we propose a novel Vessel-like Structure Rehabilitation Network (VSR-Net) to both rehabilitate subsection ruptures and improve the model calibration based on coarse vessel-like structure segmentation results. VSR-Net first constructs subsection rupture clusters via a Curvilinear Clustering Module (CCM). Then, the well-designed Curvilinear Merging Module (CMM) is applied to rehabilitate the subsection ruptures to obtain the refined vessel-like structures. Extensive experiments on six 2D/3D medical image datasets show that VSR-Net significantly outperforms state-of-the-art (SOTA) refinement segmentation methods

with lower calibration errors. Additionally, we provide quantitative analysis to explain the morphological difference between the VSR-Net’s rehabilitation results and ground truth (GT), which are smaller compared to those between SOTA methods and GT, demonstrating that our method more effectively rehabilitates vessel-like structures.

Index Terms—Vessel-like structure rehabilitation, medical image segmentation, graph convolutional network, calibration.

I. INTRODUCTION

THE morphologies of vessel-like structures (such as blood vessels, trachea, lymphatic vessels, and nerve fibres) are closely related to the pathogenesis of different diseases [1]. Accurate vessel-like structure segmentation is critical for the auxiliary diagnosis of related diseases [2], [3]. This is because morphological parameters extracted from vessel-like structure segmentation results can provide a potential reference for analyzing etiopathogenesis, significantly affecting the clinical diagnosis results. For example, the density and thickness of retinal blood vessels in the fundus are essential indicators for assessing the condition of retinal edema [4], [5] and venous occlusion [6].

In the past decades, conventional segmentation methods have widely been utilized to address vessel-like structure segmentation tasks, such as contour model-based method [9], threshold-based segmentation methods [10], and edge detection methods [11]. These methods heavily rely on manual feature extraction design and continuous threshold adjustment, which are time-consuming and fragile. Moreover, there are many subsection ruptures in vessel-like structures segmentation results based on conventional segmentation methods. The advent of deep networks and ongoing progress in medical image segmentation usher in a paradigm shift in vessel-like structure segmentation tasks. Deep network-based semantic segmentation methods [8] have simplified the segmentation process of conventional segmentation methods in an end-to-end manner. Nonetheless, they also bring subsection rupture problems in vessel-like structure segmentation results. Fig. 1(b) offers vessel-like structure segmentation results generated by U-Net [8] based on color fundus images. We see that there exists a subsection rupture problem in segmented vessel-like structures through comparisons to the two enlarged sub-regions of ground truth (as shown in Fig. 1(c)). Two possible reasons to explain this phenomenon: (1) Most deep network-based semantic segmentation methods provide category information based on pixel-wise predicted probability

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Data is available on-line at <https://github.com/YeahHighly/TIP-2024-VSR-Net>.

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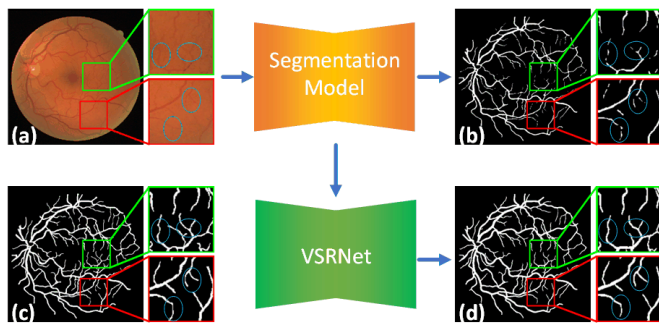


Fig. 1. (a) One representative color fundus image selected from the DRIVE [7] dataset. (b) Coarse vessel structure segmentation result generated by U-Net [8]. (c) Ground truth. (d) Vessel structure rehabilitation by VSR-Net. In the two enlarged sub-regions on the color fundus image (top left corner, marked with a blue circle), VSR-Net effectively rehabilitates continuous ruptures in U-Net's coarse vessel segmentation results.

values, unable to rectify inconsistent segmentation results around subsection fracture positions [12], [13]. (2) Inherent imaging factors like similar density distributions and blurred vessel-like morphologies also bring difficulties for existing methods to segment small vessel structures accurately. In particular, the subsection rupture problem breaks the integrity of vessel-like structures, negatively affecting the disease diagnosis results of clinicians [2].

The current mainstream approaches to improve vessel-like structure segmentation results are deep network-based refinement and topology-preserving segmentation methods. Deep network-based refinement methods have been extensively applied to refine coarse segmentation results with complex multi-scale paths and attention mechanisms [14], [15], [16], [17], [18], [19], [20], [21], [22]. However, these methods often fail to maintain spatial continuity in vessel-like structures, which inevitably breaks the overall topological integrity. The main reason is that they have limitations in repairing subsection ruptures, making it difficult to rehabilitate them back into a cohesive whole. In contrast, topology-preserving segmentation methods [23], [24], [25], [26], [27], [28] aim to maintain the connectivity and continuity of vessel-like structures by incorporating topological constraints [29], [30]. They enhance overall topological consistency and ensure that segmentation results align with the original vessel-like structure geometry [24], [26], [31]. Nevertheless, placing excessive emphasis on maintaining global topology imposes heavy constraints on the model, making it overly conservative when dealing with complex vessel-like structures. Specifically, when segmenting subtle and fragile vessel endpoints, the model tends to avoid generating aggressive predictions to preserve overall structural continuity and consistency. According to the above analysis, we found that existing methods typically excel only in a single aspect, either segmentation accuracy or morphological similarity, but struggle to achieve both effectively. Therefore, keeping an expected balance between global topological consistency and precise subsection rupture rehabilitation remains a significant challenge, especially in small vessel-like structures, e.g., subtle ruptures.

In addition, existing vessel-like structure segmentation methods are over-confident in predicted segmentation results,

increasing the risk of clinical diagnosis errors. Although expected calibration error (ECE) [32], [33] is a well-accepted metric for evaluating model calibration, it is underutilized in vessel-like structure segmentation tasks. Moreover, we find that there exist spatial interconnection relationships among subsection ruptures by observing vessel-like structure segmentation results with U-Net (as shown in Fig. 1(b)). We argue that subordinate relationships of subsection ruptures can be inferred via the end of curvilinear direction. Graph Convolutional Networks (GCNs) [34], [35] have demonstrated strong capabilities in capturing complex spatial relationships, showing great potential in improving vessel-like structure segmentation performance by rehabilitating subsection ruptures. However, the utilization of GCNs in vessel-like structure segmentation tasks, especially in repairing subsection ruptures, remains underexplored [36]. By leveraging the powerful feature representational capabilities of GCNs, it is beneficial to better capture subtle differences in small vessel-like structures, effectively rehabilitating subsection ruptures; thereby improving overall segmentation performance and topological consistency.

Motivated by the above systematic analysis, we provide a unique perspective to understand the subsection rupture problem by considering it as vessel-like structure rehabilitation rather than vessel-like structure segmentation/refinement. Here, we suggest that the vessel-like structure rehabilitation is able to keep a promising trade-off between subtle rupture rehabilitation and global topological consistency simultaneously, as well as boost the confidence calibration of the model effectively. Therefore, we propose a novel Vessel-like Structure Rehabilitation Network (VSR-Net) to rehabilitate subsection ruptures in coarse vessel-like structure segmentation results and improve the confidence calibration, as shown in Fig. 1(d). In particular, in the preparation stage, we obtain coarse vessel-like structure segmentation results with existing vessel-like structure segmentation baselines, e.g., U-Net. Next, in the rehabilitation stage, we first present a Curvilinear Clustering Module (CCM) to convert subsection ruptures into graphs and construct the clustering relationships among individual subsection ruptures via graph convolution operation, which exploits the potential of spatial interconnection relationships among subsection ruptures. Then, this paper introduces a well-designed Curvilinear Merging Module (CMM) to rehabilitate subsection ruptures in coarse segmented vessel-like structures, guided by clustering results generated by the CCM. The VSR-Net not only preserves the topological integrity well but also successfully restores subsection ruptures in vessel-like structures and improves the model calibration, overcoming the limitations of previous deep network-based refinement and topology-preserving segmentation methods.

Surprisingly, we found that existing methods have not systematically evaluated vessel-like structure rehabilitation from the structural morphology perspective, which is important for clinical diagnosis [4], [25], [28]. In this paper, we comprehensively evaluate vessel-like structure morphology rehabilitation results using metrics such as Street Mover Distance (*SMD*) [25], absolute Betti Errors (β^{err}) [28], Vascular Bifurcation Number (*VBN*) [37], Fractal Dimension (*FD*) [38], and

Vascular Tortuosity (VT) [39]. These metrics are specifically chosen under the real clinical diagnosis requirements, objectively measuring morphological differences between the prediction results and ground truth. For instance, subsection ruptures at bifurcation sites reduce the VBN, thereby increasing the risk of misdiagnosing abnormal vascular congestion.

We summarize the main contributions of this paper as follows:

- We are the first to consider the subsection rupture problem as a vessel-like structure rehabilitation task rather than a vessel-like structure segmentation task, providing novel insight into addressing this problem.
- We propose a novel Vessel-like Structure Rehabilitation Network (VSR-Net) to rehabilitate subsection ruptures and improve confidence calibration performance based on coarse vessel-like structure segmentation results, fully leveraging the potential of spatial interconnection relationships among subsection ruptures via graph clustering.
- We systematically evaluated the performance of vessel-like structure rehabilitation from the structure morphology perspective. The experimental results on 2D/3D medical image datasets show that our VSR-Net performs better in terms of vessel-like structure segmentation results, vessel-like structure rehabilitation results, and confidence calibration through comparisons to SOTA methods.
- We conduct quantitative analysis and ablation study to validate the generalization ability and robustness of our method.

II. RELATED WORK

A. Vessel-Like Structures Segmentation

Scholars have achieved significant progress in vessel-like structure segmentation tasks using 2D and 3D semantic segmentation methods. For *2D vessel-like structures segmentation*: Gu et al. proposed CENet [12] for vessel-like structure segmentation (e.g., fundus blood vessels [7]) using spatial pyramid pooling. Tan et al. introduced a retinal vessel segmentation approach [13] by applying a skeletal prior and contrastive loss to improve the precision and topological integrity of vessel-like segmentation. Li et al. introduced an attention-based network [16] that automates retinal vessel segmentation by capturing multi-scale vessel structures. Similarly, [40] incorporated attention modules and multi-channel feature fusion strategies into deep network-based semantic segmentation methods for learning salient features in curvilinear segmentation tasks. In *3D vessel-like structures segmentation*: In contrast to 2D vessel-like structures, 3D vessel-like structures exhibit indescribable intricacies in terms of textural characteristics and elongation of curvilinear segments [41], [42], [43]. Shin et al. [14] propose a method that enhances vessel segmentation by integrating GCNs with CNNs to learn both local features and global topological connectivity, improving the segmentation of complex vessel structures. Similarly, Wang et al. [31] proposed a specific-designed deep distance transform method for tubular structure

segmentation in CT scans, addressing the issue of delicate and elongated structure segmentation. Hatamizadeh et al. [44] and Liu et al. [45] leveraged the strong feature extraction capabilities of Transformers to enhance 3D medical segmentation performance. Although existing methods have achieved considerable success in vessel-like structures segmentation, they still have limitations in keeping the structural continuity and topological integrity of curvilinear segments, inevitably bringing challenging issues like subsection ruptures. To address these challenges, several deep network-based refinement and topology-preserving segmentation methods have been developed to refine coarse segmentation results or maintain topological integrity to prevent subsection ruptures and ensure continuous segmentation of vessel-like structures.

B. Deep Network-Based Refinement Segmentation

Deep network-based refinement segmentation aims to generate a more precise depiction of objects or areas within an image by refining coarse segmentation results. Jha et al. proposed DoubleUNet [17], which utilized two sequential UNet architectures to generate coarse segmentation results and refine them for accuracy improvement. Similarly, Wu et al. proposed RADCU-Net [21] to enhance retinal vessel segmentation results by integrating residual attention and dual supervision. Liu et al. developed a two-path network TP-Net [22] to improve segmentation accuracy. Other works [18], [20] followed this two-stage refinement paradigm, focusing on refining coarse segmentation outputs through iterative stages. Additionally, some methods achieved refined segmentation by designing complex attention mechanisms. For example, Wang et al. proposed EAR-Net [15], which used an error attention refining network to focus on correcting errors in a cascaded manner for retinal vessel segmentation. Li et al. introduced AGLN [19] by utilizing an attention-guided global enhancement module and a local refinement network to aggregate global semantic information. Similar works [42], [46] have also applied attention mechanisms to boost refined segmentation precision. We see that these methods have improved segmentation results through cascaded network architectures and attention mechanisms at the cost of increasing the complexity. In particular, they ignore the structural integrity maintenance of vessel-like structures and the particularity of the subsection rupture problem, causing them to address the subsection rupture problem and preserve the topological integrity poorly.

C. Graph Convolutional Network in Segmentation

Recently, GCNs [34] have been extensively applied to semantic segmentation tasks, yielding significant progress. Zhao et al. [36] introduced a graph convolution-based cross-network multiscale feature fusion method to improve vessel segmentation performance by capturing complex vessel structures. Hu et al. [47] developed a class-based dynamic graph convolution module that adaptively propagates information and performs graph reasoning between pixels of the same category. Similarly, Li et al. [48] designed a dynamic channel graph convolutional network for edge enhancement, utilizing

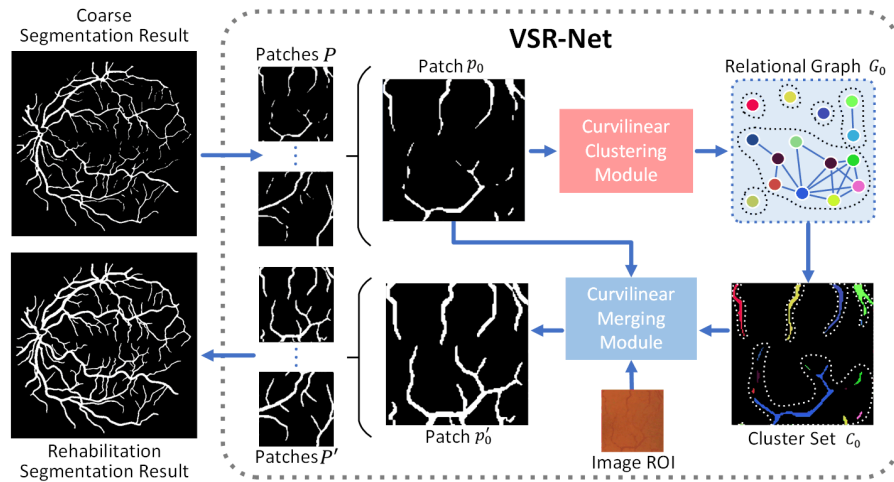


Fig. 2. Overview of the VSR-Net. (1) the coarse segmentation results are divided into individual patches P and then we input them into the Curvilinear Clustering Module (CCM). (2) CCM constructs a relationship graph G based on the vessel-like subsection ruptures in the input patch and then generates cluster set C . (3) the Curvilinear Merging Module (CMM) rehabilitate the subsection ruptures in the input patch according to cluster set C , and combines the rehabilitation results of all patches to obtain the rehabilitation segmentation result of the entire image.

dual encoders to fuse edge and spatial features. Other studies have focused on converting categories, attributes, and features into graph representations, subsequently leveraging GCNs for feature extraction and induction to boost segmentation results. GCNs offer advanced capabilities in graph-structured data feature extraction and spatial correlation analysis, but they have rarely been applied to address the specific challenges of subsection ruptures in vessel-like structure segmentation. Unlike previous methods [36], [47], [48] that mainly utilized GCNs to enhance non-Euclidean feature extraction and fusion, this paper aims to rehabilitate ruptured segments in vessel-like structures and reconstruct their continuity and integrity with the aid of GCNs, rather than solely enhancing feature attributes.

D. Topology-Preserving in Segmentation

Topology-preserving segmentation focuses on maintaining the inherent topological properties, such as connectivity and continuity of segmented structures. Stucki et al. [28] preserved the topological features of structures in topologically faithful image segmentation through induced matching of persistence barcodes. Qiu et al. [29] proposed CorSegRec to preserve full connectivity of coronary arteries from CT angiography images by preserving topology and preventing segmentation discontinuities. Zhang et al. [30] proposed a comprehensive framework to improve anatomical accuracy and keep topological integrity in coronary artery segmentation. Qi et al. [27] presented a dynamic snake convolution model to better segment tubular structures by incorporating topological geometric constraints. Other similar studies [23], [24], [25], [26] also incorporated topological constraints into the segmentation process to maintain connectivity and structural integrity, such as specialized loss functions or graph-based methods. Despite their effectiveness in preserving overall topological consistency, they often struggle to rehabilitate subsection ruptures of vessel-like structures.

E. Semantic Segmentation Calibration

Confidence calibration is to predict probability by estimating representative of true correctness likelihood, which is significant in computer-aided medical diagnosis. Over the years, several works have been proposed to tackle calibration in semantic segmentation. Zhao et al. [49] systematically studied the calibration of semantic segmentation methods and proposed a selective scaling method to improve the predicted segmentation confidence. Mehrtaash et al. [50] proposed model ensembling to improve confidence calibration of deep network-based semantic segmentation methods in medical image segmentation tasks. Ding et al. [51] presented a local temperature scaling method to extend temperature scaling to each pixel. Other methods [52], [53], such as adversarial training, domain shift, MC-dropout, and stochastic processes, are also introduced to enhance confidence calibration in semantic segmentation tasks. Unlike previous methods, we attempt to improve confidence calibration of vessel-like structure segmentation results from the structure rehabilitation perspective. Additionally, ECE [32] is employed to evaluate the confidence calibration of various methods.

III. METHOD

A. Overview of VSR-Net

In this paper, we introduce a novel Vessel-like Structure Rehabilitation Network (VSR-Net) to rehabilitate coarse vessel-like structure segmentation results brought by subsection ruptures problem, as shown in Fig. 2. Our VSR-Net comprises two main modules: Curvilinear Clustering Module (CCM) and Curvilinear Merging Module (CMM). As illustrated in Fig. 2, the general flowchart of VSR-Net is described as follows:

- **Preprocessing:** For coarse segmentation results (such as from U-Net [8]), using a sliding window to segment it into a set of patches.

- **Subsection Ruptures Clustering:** Enter each patch into the CCM module, which generates a relationship graph G and then extracts cluster set C from it. According to Fig. 2, the subsection ruptures with white dashed lines in each cluster should be rehabilitated as a complete vessel-like structure object.
- **Subsection Ruptures Rehabilitation:** The CMM rehabilitates the subsection ruptures in the patch based on cluster set C , patch image and its corresponding image ROI, and finally combines all the rehabilitation results to obtain the rehabilitation result of the entire image.

Considering the integrity of vessel-like structures, most subsection ruptures can be viewed as part of some vessel-like structures from a global perspective. Thus, inputting the entire coarse segmentation image into VSR-Net directly, may cause CMM to aggregate all subsection ruptures into a single cluster. The cluster results contradict our expectations. To address this problem, at the preparation stage, we use the sliding window strategy to divide the input image into patches $P = [p_0, p_1, \dots, p_n]$ (as shown in Fig. 2) and then input them into VSR-Net. In this paper, we set the sliding window size to 1/16 of the size of the original image in the 2D images, e.g., the length and width are 1/4 of the original image. For the 3D image, the size of the sliding window is $64 \times 64 \times 64$. The following sections will introduce CCM and CMM in detail.

B. Curvilinear Clustering Module

Fig. 3 shows an overview of the CCM. When each patch image is input into CCM, the connected components in the patch image is first obtained through connected component analysis. As shown in Fig. 3, each subsection rupture is marked with a different color to represent a connected component. The next process of CCM is as follows:

- **Curvilinear Node Modeling:** It maps the connected components into graph nodes. Each node represents a connected component, that is, a subsection rupture.
- **Node-edge Clustering:** Based on the obtained graph nodes, the Node-edge clustering algorithm is applied to cluster graph nodes, and then we merge clustering graph nodes to construct relationship graph G .
- **Graph Edge Rethinking:** Following Node-edge Clustering, this paper uses the graph edges rethinking module to extract graph features and prune the edges without connection relationships, aiming to obtain the refined relationship graph G' . According to the connection status of the nodes in G' , all nodes are split into several clusters to obtain the cluster set C with cluster separation.

1) *Curvilinear Node Modeling:* Each subsection rupture in the patch image can be viewed as a node in the graph, and the minimum distance between different subsection ruptures can be used to represent the weight of the edges for corresponding nodes. Therefore, the spatial interconnection correlation of subsection ruptures can be converted into a relationship graph G . The subsection ruptures of this relationship graph for vessel-like structure can be described as a tuple $G(N, E)$. Where N signifies the node feature set of the graph via a matrix with dimensions $|N| * L$, where $|N|$ represents the

number of graph nodes, and L represents the dimension of the node feature vector. Meanwhile, E represents the edge set within the graph. Note that constructing the relationship graph G involves converting vessel-like structure subsection ruptures into graphical representations.

In order to obtain the node feature set N of the relationship graph G , we adopt ResNet-18 as a feature extractor f_e to encode each curvilinear ROI image into feature vectors with 512. Specifically, we disentangle the curvilinear ROI image of each subsection rupture from the patch image by computing the minimum enclosing rectangle (MER) area of each connected component, then eliminating other irrelevant subsection ruptures. This operation ensures each curvilinear ROI image only contains the corresponding single subsection rupture object (as shown in Fig. 3, curvilinear node modeling for the orange dashed line area). To allow all curvilinear ROI images to be fed into the feature extractor f_e in parallel, 2D images are resized to 224×224 before input, while 3D voxels are resized to $64 \times 64 \times 64$ [54]. The process to obtain the feature vector of each subsection rupture is formulated as follows:

$$N = \{n_1, n_2, \dots, n_{|N|}\} \\ = \{f_e(v_1), f_e(v_2), \dots, f_e(v_{|N|})\}, \quad (1)$$

where $\{n_1, n_2, \dots, n_{|N|}\}$ are the node features with length $L = 512$, each node feature n is extracted from the curvilinear ROI image v using the feature extractor f_e . The feature vector of each node represents the corresponding features of the subsection rupture it is associated with.

2) *Node-Edge Clustering:* Next, adopting the Top- K nearest subsection ruptures as neighbor nodes, as shown in Fig. 4. Here, we take $K = 10$ as an example, the target node (red node) connects the ten closest nodes based on the Top- K nearest results. Subsequently, setting a distance threshold τ , then the edges with a distance greater than τ will be trimmed; finally, the remaining edges will be used as the useful edges of the node. As shown in Fig. 4, taking $\tau = 30$ as an example, the target node (red node) prunes the two nearest neighbors whose shortest distance exceeds the threshold τ . Based on this, the edges of each node are clustered by the node-edge clustering method. Then, the clustering results of each node are combined to generate the complete graph edge set E . If the edge between two nodes is not pruned, it indicates that the edge is valid; that is, there is a connection between the two nodes. After determining the graph's complete graph edge set E , the edges' weight is normalized by relationship graph G . We set the total edge number to Z and the weight E_i for the i edge as:

$$D_{A,B} = \min_{a \in A, b \in B} f_{dis}(a, b), \quad (2)$$

where f_{dis} represents the euclidean distance metric function, the a and b are points of subsection ruptures A and B , respectively, indicating the shortest distance between the two subsection ruptures.

As shown in Fig. 4, take $K = 10$ as an example, the target node (red node) connects the ten closest nodes in

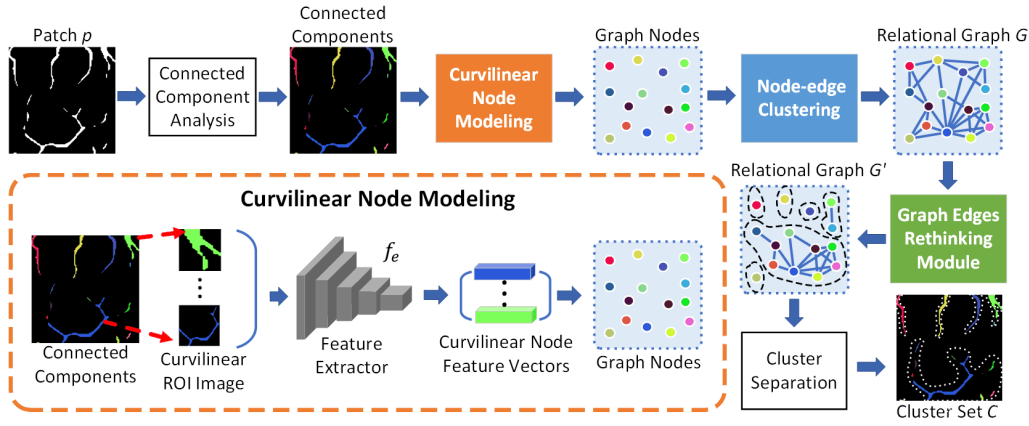


Fig. 3. Overview of the Curvilinear Clustering Module (CCM) process. CCM first models the subsection rupture of the patch image p into the nodes of the graph through curvilinear node modeling; Then, it clusters the edges of the nodes through the Node-edge clustering algorithm to obtain the relationship graph G ; Finally, the refined relationship graph G' is obtained by pruning the edges without a connection relationship through the graph edge rethinking module, and dividing them through cluster separation to obtain the cluster set C .

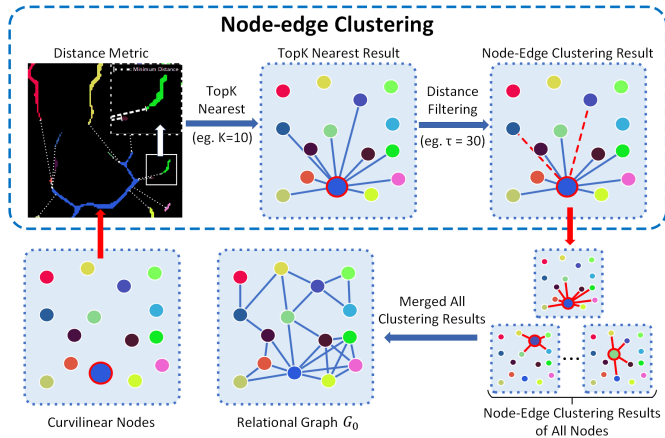


Fig. 4. Overview of Node-edge Clustering process. For each curvilinear node clustering needs processing distance metric, Top-K nearest neighbor search and distance threshold filtering respectively. Then, all the clustering results are combined to get the edge set E of the relationship graph G .

the TopK nearest result. Subsequently, set a distance threshold τ , and the edges with a distance greater than τ will be trimmed, and the remaining edges will be used as the effective edges of the node. As shown in Fig. 4, take $\tau = 30$ as an example, the target node (red node) prunes the two nearest neighbors whose shortest distance exceeds the threshold τ . Based on this, the edges of each node are clustered by node-edge clustering method. Then, the clustering results of each node are combined to get the complete graph edge set E . If the edge between two nodes is not pruned, it indicates that the edge is valid, indicating that there is a connection between the two nodes. Algorithm 1 summarizes the clustering details of node-edge clustering. After determining the graph's complete graph edge set E , the edges' weight is normalized by relationship graph G . Set that there are Z edges in total, and the weight E_i of the i edge is:

$$E_i = \frac{e^{\frac{1}{D_i}}}{\sum_{z=1}^Z e^{\frac{1}{D_z}}}, \quad (3)$$

Algorithm 1 Node-Edges Search Algorithm

Input: Vascular image set $\{v\}$. Initialised parameters of neighbour matrix set $\{w\}$. Hyperparameter of maximum neighbours number K and distance threshold τ .

Output: The parameters of the neighbour matrix set $\{w\}$.

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1: for  $i$  in range( len( $v$ ) ) do
2:   Initialized temp list,  $T = []$ .
3:   for  $j$  in range( len( $v$ ) ) do
4:     if  $i \neq j$  then
5:       Compute the minimum distance  $D_{v_i, v_j}$ .
6:       Append the  $D_{v_i, v_j}$  to  $T$ .
7:   Search the first  $K$  minimum distance value indexes in  $T$ , get the indexes set  $r$ , the  $\text{len}(r) \leq K$ .
8:   for  $j$  in range( len( $v$ ) ) do
9:     if  $D_{v_i, v_j} < \tau$  and  $j \in r$  then.
10:       $w_{i,j} = T_j$ , connection.
11:   else
12:      $w_{i,j} = \infty$ , non-connection.
13: return  $\{w\}$ 

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where, D_i represents the original minimum distance of the i edge. The Eq. (3) adopts the Softmax function to normalize the weight of the edge. If the shortest distance between two nodes larger, the weight of the edge between them should be smaller. Thus, the relationship graph G is composed of the set of graph edges set $E = \{e_1, e_2, \dots, e_p\}$ and the node feature set $N = \{n_1, n_2, \dots, n_{|N|}\}$.

3) *Graph Edges Rethinking Module*: Graph Edge Rethinking Module (GERM) considers the interconnection relationship among the subsection ruptures, represented by each node in the relationship graph G . GERM also prunes the edges without connection relationships in G . The feature extraction process in the GERM is implemented by GCN, which is formulated as follows:

$$h_i^{l+1} = \sigma \left(\sum_{j \in N_i} \frac{1}{c_{ij}} h_j^l w^l + b^l \right), \quad (4)$$

where h_j^l is the feature of node j at the l layer, σ is the LeakReLU function. c_{ij} is a normalized factor, w^l and b^l are learnable weights and bias of the l layer. N_i is the set of neighboring nodes of node i , including its own nodes. Eq. (6) describes the operation flow of layer $l + 1$ in GCN.

Moreover, to dynamically adjust the relative importance of neighbours of each node, we introduce the attention mechanism [55] in GCN by calculating the attention coefficient of the current node and its neighbors. Then, the node features are weighted and summed according to the calculated attention coefficient. The attention mechanism in the GCN layer is computed as follows:

$$h_i^{l+1} = \sigma\left(\frac{1}{K} \sum_{k=1}^K \sum_{j \in N_i} \frac{1}{c_{ij}} \alpha_{ij}^k h_j^l W^{l,k} + b^l\right), \quad (5)$$

where α_{ij}^k is the normalised attention coefficients computed by the k -th attention mechanism. After the graph feature representation extraction of the GCN layer, the end of the GERM is the FC layer for classifying edge categories. Specifically, the categories of each edge in the refined relationship graph G' is predicted. The two-node features connected by each edge are spliced and then input to the FC Layer to get the prediction result in $y_{E_i} = \Theta(F_{fc}([h_a||h_b]))$, Θ is Sigmod function. When the predicted result is 0, this edge will be deleted. In contrast, this edge will remain when the result is 1.

Finally, as for the design of GERM loss, in addition to the classification loss of edge prediction, regularisation is also adopted to enrich the diversity of multi-head attention learning:

$$L_{gcn} = \frac{1}{T} \sum_{t=1}^T L_{bce}(y'_{E_t}, y_{E_t}) + L_r(O), \quad (6)$$

$$L_r(O) = -\frac{1}{L * H^2} \sum_{l=1}^L \sum_{i=1}^H \sum_{j=1}^H \frac{O_{l,i} \cdot O_{l,j}}{||O_{l,i}|| ||O_{l,j}||}, \quad (7)$$

where L_{bce} is the binary cross-entropy loss as edge classification loss. y_{E_t} is the predicted result of GERM, and y'_{E_t} is the label of the edge. $L_r(O)$ is attention regularization, $O_{l,i}$ represents the output of i -th GCN attention at layer l . Specifically, Eq. (7) calculates the cosine similarity $\cos(\cdot)$ between the same layer attention outputs pair through the normalised dot product. This regularisation directly applies to the outputs of each attention head by maximising the difference among them. The clustering process in the CCM module effectively leverages spatial interconnection relationships between subsection ruptures, allowing the subsequent CMM module to determine which subsection ruptures can be merged into a cohesive whole during rehabilitation, thereby ensuring an expected topological integrity in the final outputs.

C. Curvilinear Merging Module

According the process of CCM, we obtain the cluster set C . To merge the subsection ruptures of each cluster into a complete subsection, we propose the curvilinear merging module (CMM). The process of merging the subsection ruptures into an entire subsection via the proposed CMM, as shown in Fig. 5. In each cluster, CMM initially captures the minimum

enclosing rectangle area around the subsection ruptures. Subsequently, it eliminates any other subsection ruptures located in this area that do not belong to this cluster. CMM cuts and pads the image ROI and patch respectively, for producing the mapping sets corresponding to the clusters. To rehabilitate in CMM concurrently, the paired images (ROI mapping and cluster mapping) are resized to match the size of the input patch, which is 1/16 of the original image size for 2D images, and $64 \times 64 \times 64$ for 3D voxels [54]. These two mapping sets are fed into two feature encoder modules, denoted as f_{m1} and f_{m2} , performing feature extraction on the ROI area R and the original image I , respectively. The feature representations obtained from these two encoder modules are then integrated with the fusion step. Subsequently, fused feature representations are passed through the decoder, represented as f_d , to generate the cluster merging result.

The encoder and decoder modules in CMM are implemented based on the light UNet architecture [56]. To enhance vessel-like structure rehabilitation performance, we delete two sets of convolutional layers and downsampling layers at the high stage of both encoder and decoder, resulting in a downsampling ratio change from 32 to 8. This adjustment aims to preserve more shallow features. Additionally, skip connection mechanisms have been introduced between the encoder and decoder to transmit texture and shape information across different stages. Finally, all the cluster merging results are mapped according to their relative positions and combined into patch-based vessel-like structure rehabilitation results.

Considering curvilinears only occupy a small area of the patch image, the pixel cross-entropy loss and Dice loss cannot describe the subtle differences in the vessel-like segmentation well. The CMM loss is proposed as follows:

$$L_{cmm} = L_{ce}(R', \hat{R}') + L_{pl}(R', \hat{R}') + L_{cc}(R'), \quad (8)$$

where R' is the result of the revised merged model, and $\hat{R}'$ is the ground truth. L_{ce} is the cross-entropy loss responsible for guiding at the beginning of model training. L_{pl} is the polygon line loss, which mainly measures the polyline similarity between R' and $\hat{R}'$:

$$L_{pl} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \min_{q \in Q} ||s - q_i||_2 + \frac{1}{|S|} \sum_{j=1}^{|S|} \min_{s \in S} ||q - s_j||_2, \quad (9)$$

where S and Q are pixel points sets of subsections in R' and $\hat{R}'$ respectively.

L_{pl} measures the distance difference of subsection pixels between S and Q . In the CMM, we expect all subsection ruptures in R' to be merged into a more complete subsection. When the number of connected components in R' is greater than 1, indicating that there are subsection ruptures or subsections that still have not been merged. To address this problem, we propose connected components loss $L_{cc}: L_{cc} = 1 - \frac{1}{C}$, where C is the number of connected components of R' . When the number of connected components of R' decreases, the loss will also decrease, encouraging the CMM to segment the subsections together to obtain promising vessel-like structure rehabilitation results. Benefiting from the ingenious loss design

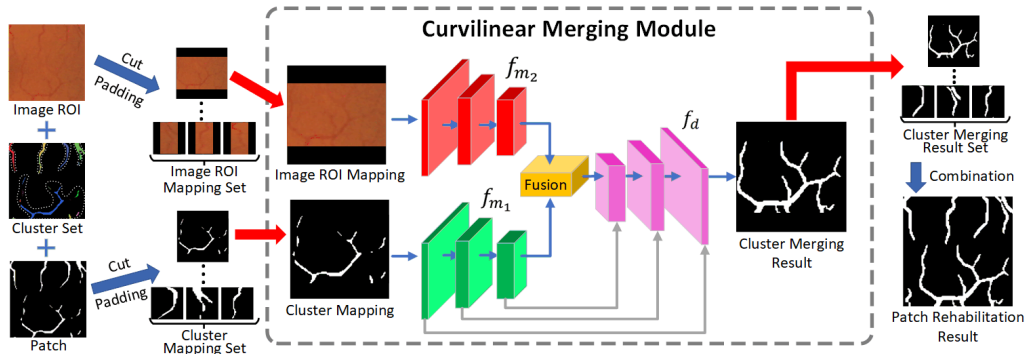


Fig. 5. Overview of the Curvilinear Merging Module (CMM) process. The CMM cuts and pads the mapping sets of the image ROI and Patch based on the cluster set. Subsequently, these sets are paired and input into the CMM module to derive cluster merging results. Finally, the entirety of cluster merging results is consolidated into the patch rehabilitation results.

TABLE I

INFORMATION FROM FIVE PUBLIC DATASETS

Dataset	Training-test	Object	Modality
DRIVE	20-20	Fundus vessels	Fundus images
OCTA-500	240-60	Fundus vessels	OCTA
CORN-1	1176-340	Corneal nerve	CCM images
ATM	200-99	Pulmonary trachea	CT
MIDAS-I	30-20	Intracranial artery	MRI
ImageCAS	800-200	Coronary artery	CTA

in the CMM module, the rehabilitation process can simultaneously preserve the detailed features of vessel-like structures. Specifically, the connected components loss L_{cc} encourages the CMM module to merge subsection ruptures within the same cluster into a cohesive whole, ensuring the topological integrity of the vessel-like structures. Additionally, the polygon line loss L_{pl} further promotes precise segmentation of fragile vessel-like structure edges. Therefore, fully exploring the merits of the CCM and CMM modules guarantees that VSR-Net effectively balances global topological consistency with the need for accurate subsection rupture rehabilitation, particularly in small vessel-like structures, achieving a balance between topological integrity and fine segmentation.

IV. EXPERIMENTS

We first introduce datasets in Sec. IV-A. Then, implementation details and evaluation metrics are listed in Sec. IV-B. Followed by ablation studies and performance comparisons with SOTA methods in Sec. IV-C and IV-D. Finally, we provide quantitative analysis to demonstrate the robustness of our method in Sec. IV-E.

A. Datasets

In this paper, we adopt five 2D/3D medical image datasets to verify the effectiveness of our method. The 2D medical image datasets include DRIVE, OCTA-500, and CORN-1. The 3D medical image datasets include ATM, MIDAS-I, and ImageCAS, as listed in Table I.

1) 2D Vessel-Like Segmentation Datasets:

DRIVE [7].¹ It contains 40 pairs of fundus images and corresponding labels of vessel-like object segmentation. The size of each image in the dataset is 565×584 .

¹<http://www.isi.uu.nl/Research/Databases/DRIVE/>

OCTA-500 [57].² It contains OCTA images of the eyes of 300 patients with $6\text{mm} \times 6\text{mm}$ FOV. Only one image for each subject's eye, ensuring diversity and avoiding similar data. The size of each image in the dataset is 400×400 .

CORN-1 [58].³ It contains 1516 CCM images of subbasal corneal epithelium using a Heidelberg Retina Tomograph equipped with a Rostock Cornea Module (HRT-III) microscope. Each image has a resolution of 384×384 pixels covering a FOV of $384 \times 384 \mu\text{m}^2$ with manual annotation.

2) 3D Vessel-Like Segmentation Datasets:

ATM [41].⁴ The dataset contains 299 CT scans from multi-sites. Three radiologists carefully label the airway tree structures with more than five years of professional experience. Each chest CT scan consisted of various slices, ranging from 157 to 1125, with a slice thickness of 0.450–1.000 mm. The axial size of all slices is 512×512 pixels with a spatial resolution of 0.500–0.919 mm.

MIDAS-I [42].⁵ It is a subset of MIDAS, including 50 manual annotations of intracranial arterial in MRA. MIDAS dataset is a public cerebrovascular dataset that contains a total of 100 MRA volumes that were acquired from healthy volunteers aged 18 to 60+ years. A core region containing intracranial arteries with dimensions of $224 \times 208 \times 64$ was cropped from each raw data to minimise interference with other brain regions.

ImageCAS [43].⁶ The dataset consists of 1,000 chest CTA scans collected from multiple medical centers. Four radiologists with over six years of experience meticulously annotated the coronary arteries. The CT scans have slice counts ranging from 150 to 1,200, with a slice thickness between 0.5 and 1.0 mm. The in-plane resolution of the images is 512×512 pixels, with a spatial resolution between 0.6 and 0.9 mm.

B. Experimental Settings

1) Implementation Details: We implemented VSR-Net and comparable methods using the PyTorch platform.

²<https://ieee-dataport.org/open-access/octa-500>

³<https://imed.nimte.ac.cn/CORN.html>

⁴<https://atm22.grand-challenge.org/>

⁵<https://public.kitware.com/Wiki/TubeTK/Data>

⁶<https://www.kaggle.com/datasets/xiaoweixumedicallai/imagecas>

The Stochastic Gradient Descent (SGD) optimizer is used with a momentum of 0.99 and a weight decay of 0.0001. Training is conducted for 100 epochs. The batch size of CCM and CMM was set to 4 and 10, respectively. The learning rates of CCM and CMM are 10^{-3} and 10^{-4} , respectively. The batch size and learning rate for other segmentation networks are 4 and 10^{-4} in training. Standard data augmentation methods are applied during training, such as random scaling (from 0.5 to 2.1) and random flipping. It is worth noting that the data size input to the model during inference is the same as during the training phase; thus, we can execute all patches for the same image in parallel. All experiments are performed using two GPUs (Tesla V100).

2) *Evaluation Metrics*: In this paper, we evaluate the general performance of our method from four perspectives, both qualitatively and quantitatively: segmentation metrics, vessel-like structure rehabilitation metrics, confidence calibration metrics, and computational efficiency.

- Segmentation metrics. Three commonly used evaluation measures are adopted: Pixel Accuracy (PA), Dice coefficient ($Dice$), and Jaccard coefficient ($Jacc$).
- Vessel-like structure rehabilitation metrics. To evaluate vessel-like morphology rehabilitation, we calculate the differences in Vascular Bifurcation Number (VBN), Fractal Dimension (FD), and Vascular Tortuosity (VT). Additionally, we incorporate the mean of absolute Betti Errors (β^{err}) [28] and Street Mover Distance (SMD) [25] to better capture morphological and typological differences.
- Confidence calibration metrics. The widely accepted expected calibration error (ECE) [32] is utilized to measure the calibration of the network. Following the previous work [33], ECE is calculated in a pixel-wise manner, defined as:

$$ECE = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (10)$$

where B_m represents the set of pixels with predicted confidence values in the m th bin, $\text{acc}(B_m)$ is the accuracy of those pixels, $\text{conf}(B_m)$ denotes average confidence, and n is the total number of pixels. It is worth noting that most vessel-like structure segmentation methods obtain low ECE values on large vessel-like structures and high ECE values on small vessel-like structures. Consequently, the differences in ECE can effectively reflect the confidence levels of the model in rehabilitating subsection ruptures.

- Computational efficiency. We evaluate computational efficiency using Parameters ($P(M)$), FLOPs ($P(G)$), and FPS to measure model complexity, computational cost, and inference speed, respectively.

C. Ablation Studies

In this section, we perform ablation experiments to validate the effectiveness of CCM and CMM modules in VSR-Net, including: 1) hyperparameter selection of the CCM module, and 2) evaluation of the effects of each component in the CMM module. In the preparation stage, we adopt UNet [8] as the

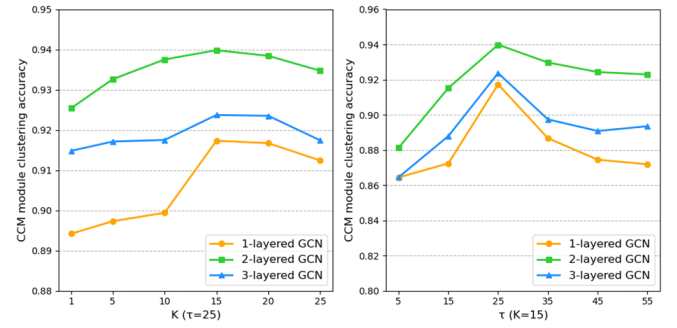


Fig. 6. The performance of CMM when node nearest neighbours TopK K , the distance threshold τ , and the number of GCN layers taking different values on DRIVE [7] test set. When $K=15$, $\tau=25$, and GCN layers=2, the CCM module builds the highest-quality performance.

TABLE II
ABLATION STUDY OF CMM ON DRIVE [7] TEST SET

$\rightarrow_{skip}^{f_{m2}}$	$\rightarrow_{skip}^{f_{m1}}$	L_{pl}	L_{cc}	PA	$Dice$	$Jaccard$
✓				0.9504	0.8415	0.7090
	✓			0.9505	0.8416	0.7092
✓	✓			0.9548	0.8503	0.7250
✓	✓	✓		0.9566	0.8579	0.7321
✓	✓	✓	✓	0.9591	0.8656	0.7434
✓	✓	✓	✓	0.9621	0.8790	0.7629
	✓	✓	✓	0.9620	0.8788	0.7625

baseline to generate coarse segmentation results based on the DRIVE dataset.

1) *Hyperparameter Selection in CCM*: We test the effects of different node nearest neighbors top-K (K) value settings on the CCM module, pruning distance threshold (τ) and the depth l of GCN layers. Fig. 6 offers the performance comparisons of different hyperparameter selections: τ and l . According to Fig. 6(Left), the CCM achieves the best performance with $K = 15$ when $\tau = 25$ pixels is fixed. Fig. 6(Right) presents the performance comparisons of different τ values in the CMM when $K = 15$ is fixed. It can be observed that CCM performs better with $\tau = 25$ than other τ value settings. In addition, we also see that GCN obtains the best performance when the depth l of GCN is 2 than other depth of layer setting, as shown in Fig. 6. Overall, the hyperparameter selections of K , τ , and l are crucial for the performance of CCM in clustering the graph G . When $K = 15$, $\tau = 25$, and $l = 2$, CCM obtains the highest performance, and we will adopt these settings for the following experiments.

2) *Effects of Each Component in CMM*: Table II illustrates the results of each component in the CMM: skip connections $\rightarrow_{skip}^{f_{m2}}$ and $\rightarrow_{skip}^{f_{m1}}$, polygon line loss (L_{pl}), and connected components loss (L_{cc}). While most components contribute to meaningful performance improvements, the skip connection from f_{m2} shows minimal or even negative impact. This could be due to f_{m2} introducing artifacts from the raw image, which may hinder the convergence of the CMM module. Therefore, in the final structure of CMM, we retained only the other three components ($\rightarrow_{skip}^{f_{m1}}$, L_{pl} , and L_{cc}) for the subsequent experiments. By excluding $\rightarrow_{skip}^{f_{m2}}$, we also improve the computational efficiency of the CMM.

TABLE III

COMPARISON OF STATE-OF-THE-ART METHODS WITH VSR-Net ON THE DRIVE [7] TEST SET ACROSS SEGMENTATION, VESSEL REHABILITATION, CONFIDENCE CALIBRATION, AND COMPUTATIONAL EFFICIENCY METRICS

Model	Year	$PA \uparrow$	$Dice \uparrow$	$Jacc \uparrow$	$VBN \downarrow$	$FD \downarrow$	$VT \downarrow$	$\beta^{err} \downarrow$	$SMD \downarrow$	$ECE \downarrow$	$P(M) \downarrow$	$F(G) \downarrow$	$FPS \uparrow$
DeepGraphSeg [14]	2019	0.9592	0.8643	0.7425	0.2921	0.5424	0.5947	4.4205	0.0179	0.0678	4.11	10.93	95
EAR-Net [15]	2021	0.9609	0.8719	0.7503	0.2851	0.5352	0.5284	3.9492	0.0163	0.0613	9.81	24.52	93
FA-Net [16]	2021	0.9607	0.8708	0.7499	0.2867	0.5372	0.5289	4.0782	0.0162	0.0615	10.81	25.93	73
DoubleUNet [17]	2022	0.9565	0.8502	0.7240	0.3669	0.5890	0.6641	4.0056	0.0185	0.0679	13.81	28.51	42
RFNet [18]	2022	0.9571	0.8539	0.7259	0.3338	0.5816	0.6395	4.1286	0.0184	0.0765	5.53	17.61	81
AGLN [19]	2023	0.9580	0.8571	0.7370	0.3117	0.5656	0.5855	4.4049	0.0180	0.0512	3.71	9.81	127
EN-UNet [20]	2023	0.9613	0.8748	0.7548	0.2802	0.5301	0.5202	3.6408	0.0160	0.0603	8.22	19.83	63
RADCU-Net [21]	2023	0.9654	0.8849	0.7599	0.1751	0.3842	0.4732	3.7869	0.0154	0.0335	6.31	14.81	67
TP-Net [22]	2023	0.9663	0.8873	0.7613	0.1737	0.3817	0.4704	3.8382	0.0152	0.0335	6.93	16.73	61
TPDI [23]	2019	0.9573	0.8598	0.7398	0.2986	0.5501	0.5885	5.0235	0.0181	0.0721	19.44	6.81	101
JTFN [24]	2021	0.9599	0.8689	0.7479	0.2901	0.5401	0.5319	4.6296	0.0168	0.0623	9.51	23.81	85
clDice [25]	2021	0.9604	0.8698	0.7489	0.2882	0.5381	0.5298	4.2057	0.0166	0.0617	9.31	23.51	87
TPLS [26]	2022	0.9582	0.8582	0.7360	0.3092	0.5612	0.5788	4.4781	0.0182	0.0701	6.41	12.53	105
DSCNet [27]	2023	0.9601	0.8679	0.7519	0.2907	0.5441	0.5456	3.8904	0.0175	0.0632	11.83	27.53	53
IMPB [28]	2023	0.9658	0.8868	0.7608	0.1742	0.3821	0.4712	3.9861	0.0152	0.0334	7.51	17.23	57
VSR-Net [†]	2024	0.9650	0.8864	0.7606	0.1769	0.3834	0.4724	4.0056	0.0157	0.0337	28.34	7.16	91
VSR-Net [‡]	2024	0.9793	0.8916	0.7662	0.1635	0.2237	0.4098	3.6882	0.0149	0.0281	26.49	6.27	109

* VSR-Net[†] and VSR-Net[‡] are the rehabilitation results based on the coarse segmentation results of CENet [12] and SKelcon-Net [13], respectively.

D. Comparisons With State-of-the-Arts

This section compares our VSR-Net with state-of-the-art (SOTA) refinement segmentation methods on several 2D and 3D vessel-like structure segmentation datasets. Specifically, we adopt the following methods: (1) 2D deep network-based refinement segmentation methods: DeepGraphSeg [14], EAR-Net [15], FA-Net [16], Double-UNet [17], RFNet [18], AGLN [19], EN-UNet [20], RADCU-Net [21], and TP-Net [22]; (2) 2D topology-preserving segmentation methods: TPDI [23], JTFN [24], clDice [25], TPLS [26], DSCNet [27] and IMPB [28]; (3) 3D deep network-based refinement segmentation methods: DeepGraphSeg [14], DeepDT [31], DARN [59], EfficientSegNet [60], ERNet [42], GraphConvSeg [36], CPM [46], and UR-CarA-Net [61]; (4) 3D topology-preserving segmentation methods: CorSegRec [29], DSCNet [27], and ATPF [30].

For our VSR-Net, we utilize CENet [12] and SKelcon-Net [13] to generate coarse 2D vessel-like structure segmentation results; nnUNet [62] and Swin UNETR [45] are applied to generate coarse 3D vessel-like structure segmentation results. Then, VSR-Net dynamically rehabilitates the subsection ruptures based on the coarse segmentation results.

1) *Performance Comparison on 2D Vessel-Like Structure Datasets*: Tables III, IV and V present a comparison of our VSR-Net with other competing methods on 2D vessel-like structure segmentation datasets, focusing on segmentation results, vessel structure rehabilitation, confidence calibration, and computational efficiency. We conclude as follows: (1) VSR-Net obtains gains of Dice ranging from 0.67% to 2.08% over attention-based refinement methods e.g., EAR-Net [15], FA-Net [16], and RADCU-Net [21] on the DRIVE [7] test set. Compared to topology-based methods like clDice [25], DSCNet [27], and IMPB [28], VSR-Net achieves Dice improvements of 0.48% to 2.37%. Additionally, compared to JTFN [24], combining refinement segmentation and topological constraints, VSR-Net improves PA and Jaccard by 1.94% and 1.83%, respectively. (2) For

vessel-like structure rehabilitation, our method significantly reduces VBN by a range of 0.0107 to 0.1482, outperforming other methods. Furthermore, VSR-Net achieves the best results in β^{err} and SMD under SKelcon-Net [13]. (3) For confidence calibration, VSR-Net outperforms other SOTA methods by reducing ECE from 0.0337 to 0.0281. (4) Regarding computational efficiency, VSR-Net's FPS is influenced by the structure of the preceding coarse segmentation model. It achieves an overall FPS of 109, which is still sufficient for clinical diagnostic support and is superior to most SOTA methods. Similarly, VSR-Net achieves the best performance across four evaluation metrics on the OCTA-500 [57] and CORN-1 [58] test sets. Overall, the experimental results in Tables III, IV, V demonstrate the advantages of VSR-Net over existing deep network-based refinement segmentation or topology-constrained methods in repairing subsection ruptures and improving segmentation quality, confidence calibration, and morphological similarity.

Next, we selected one representative sample from three different 2D vessel-like structures for visual comparisons of 2D vessel-like structure rehabilitation in terms of maintaining topological integrity and overall segmentation quality, as shown in Figure 7. Two topology-preserving methods (IMPB [28] and DSCNet [27]) and a deep network-based refinement segmentation method (TP-Net [22]) adopted. We observe: (1) VSR-Net outperforms other topology-preserving SOTA methods in rehabilitate subsection ruptures. Although the other methods can prevent dense subsection ruptures through morphological constraints and continuity modeling, they still struggle with subsection ruptures rehabilitation. (2) In cases of severe rupture, other SOTA methods fail to reconnect and rehabilitate the subsection ruptures into a cohesive whole. In contrast, VSR-Net can establish the connections between subsection ruptures, integrating them into a unified structure. (3) Regardless of variations in the thickness of the vessel-like structures, VSR-Net effectively rehabilitates and finely segments these structures, resulting in segmentation details that closely align with manually

TABLE IV
COMPARISON OF STATE-OF-THE-ART METHODS WITH VSR-NET ON THE OCTA-500 [57] TEST SET ACROSS SEGMENTATION, VESSEL REHABILITATION, CONFIDENCE CALIBRATION, AND COMPUTATIONAL EFFICIENCY METRICS

Model	Year	$PA \uparrow$	$Dice \uparrow$	$Jacc \uparrow$	$VBN \downarrow$	$FD \downarrow$	$VT \downarrow$	$\beta^{err} \downarrow$	$SMD \downarrow$	$ECE \downarrow$	$FPS \uparrow$
DeepGraphSeg [14]	2019	0.9844	0.8668	0.7610	0.2171	0.4056	0.3286	3.7167	0.0151	0.0736	124
EAR-Net [15]	2021	0.9847	0.8712	0.7598	0.2091	0.3951	0.3178	3.6675	0.0148	0.0637	115
FA-Net [16]	2021	0.9845	0.8710	0.7595	0.2114	0.3968	0.3204	3.7134	0.0148	0.0645	95
DoubleUNet [17]	2022	0.9755	0.8590	0.7504	0.2907	0.4894	0.4243	4.3356	0.0154	0.0790	53
RFNet [18]	2022	0.9799	0.8643	0.7527	0.2671	0.4757	0.3963	4.2867	0.0153	0.1195	108
AGLN [19]	2023	0.9824	0.8654	0.7533	0.2867	0.4076	0.3581	4.0023	0.0153	0.0929	169
EN-UNet [20]	2023	0.9856	0.8718	0.7647	0.2160	0.3388	0.3109	3.5496	0.0147	0.0824	83
RADCU-Net [21]	2023	0.9866	0.8750	0.7650	0.1824	0.3510	0.3005	3.5439	0.0144	0.0398	82
TP-Net [22]	2023	0.9871	0.8764	0.7667	0.1791	0.3478	0.2970	3.5235	0.0143	0.0395	79
TPDI [23]	2019	0.9829	0.8620	0.7556	0.2234	0.4128	0.3350	3.8685	0.0152	0.0778	110
JTFN [24]	2021	0.9838	0.8690	0.7576	0.2152	0.4003	0.3245	3.7104	0.0150	0.0652	107
clDice [25]	2021	0.9842	0.8700	0.7585	0.2128	0.3982	0.3217	3.6972	0.0149	0.0648	114
TPLS [26]	2022	0.9804	0.8651	0.7540	0.2635	0.4702	0.3921	4.1247	0.0153	0.0754	119
DSCNet [27]	2023	0.9839	0.8687	0.7602	0.2183	0.4021	0.3280	3.7203	0.0152	0.0664	70
IMPB [28]	2023	0.9870	0.8761	0.7665	0.1802	0.3491	0.2983	3.3981	0.0143	0.0394	72
VSR-Net [†]	2024	0.9875	0.8770	0.7671	0.1785	0.3484	0.2981	3.4623	0.0143	0.0397	106
VSR-Net [‡]	2024	0.9881	0.8767	0.7809	0.1574	0.2508	0.2038	3.3945	0.0141	0.0289	129

* VSR-Net[†] and VSR-Net[‡] are the rehabilitation results based on the coarse segmentation results of CENet [12] and SKelcon-Net [13], respectively.

TABLE V
COMPARISON OF STATE-OF-THE-ART METHODS WITH VSR-NET ON THE CORN-1 [58] TEST SET ACROSS SEGMENTATION, VESSEL REHABILITATION, CONFIDENCE CALIBRATION, AND COMPUTATIONAL EFFICIENCY METRICS

Model	Year	$PA \uparrow$	$Dice \uparrow$	$Jacc \uparrow$	$VBN \downarrow$	$FD \downarrow$	$VT \downarrow$	$\beta^{err} \downarrow$	$SMD \downarrow$	$ECE \downarrow$	$FPS \uparrow$
DeepGraphSeg [14]	2019	0.9562	0.8634	0.7571	0.3943	0.5292	0.4743	3.5689	0.0157	0.0653	151
EAR-Net [15]	2021	0.9545	0.8642	0.7569	0.3952	0.5289	0.4731	2.9943	0.0157	0.0376	148
FA-Net [16]	2021	0.9559	0.8640	0.7565	0.3954	0.5290	0.4737	3.7143	0.0157	0.0668	118
DoubleUNet [17]	2022	0.9492	0.8585	0.7507	0.5120	0.5644	0.5971	4.3361	0.0160	0.0907	68
RFNet [18]	2022	0.9528	0.8600	0.7531	0.5117	0.5346	0.5853	3.9021	0.0159	0.0522	130
AGLN [19]	2023	0.9531	0.8604	0.7548	0.4094	0.5302	0.5407	3.2948	0.0158	0.0724	209
EN-UNet [20]	2023	0.9566	0.8690	0.7604	0.3448	0.5273	0.4724	3.5748	0.0156	0.0983	100
RADCU-Net [21]	2023	0.9578	0.8701	0.7630	0.3159	0.5004	0.4347	3.5007	0.0144	0.0585	107
TP-Net [22]	2023	0.9585	0.8730	0.7660	0.3110	0.4968	0.4304	2.9640	0.0134	0.0375	99
TPDI [23]	2019	0.9548	0.8601	0.7536	0.4035	0.5348	0.4821	3.0870	0.0158	0.0375	168
JTFN [24]	2021	0.9551	0.8629	0.7554	0.3987	0.5301	0.4759	3.0171	0.0157	0.0379	141
clDice [25]	2021	0.9556	0.8635	0.7560	0.3968	0.5295	0.4745	3.0369	0.0157	0.0377	137
TPLS [26]	2022	0.9534	0.8610	0.7545	0.4087	0.5322	0.4865	2.9951	0.0158	0.0378	170
DSCNet [27]	2023	0.9552	0.8618	0.7550	0.3991	0.5305	0.4762	3.0211	0.0157	0.0378	83
IMPB [28]	2023	0.9582	0.8725	0.7653	0.3127	0.4981	0.4315	3.0437	0.0134	0.0376	92
VSR-Net [†]	2024	0.9657	0.8724	0.7678	0.2316	0.4964	0.3805	3.0861	0.0136	0.0388	158
VSR-Net [‡]	2024	0.9682	0.8754	0.7705	0.2204	0.4821	0.3628	3.0782	0.0128	0.0371	184

VSR-Net[†] and VSR-Net[‡] are the rehabilitation results based on the coarse segmentation results of CENet [12] and SKelcon-Net [13], respectively.

annotated results. (4) The comparison of visual rehabilitation results demonstrates that our VSR-Net not only achieves excellent segmentation results but also effectively rehabilitates subsection ruptures in vessel-like structures through a vessel-like structure rehabilitation perspective, consistent with expectations.

2) *Performance Comparison on 3D Vessel-Like Structure Segmentation Datasets:* Tables VI, VII, and VIII provide a comparison of our VSR-Net with competing methods on 3D vessel structure datasets. It can be observed that our VSR-Net consistently outperforms similar 3D refinement or topology-based segmentation methods. First, in terms of segmentation accuracy, the VSR-Net based on Swin UNETR shows improvements of 2.01% in Dice and 2.45% in Jaccard indices. Compared to topology-constrained methods like CorSegRec [29] and DSCNet [27], the improvements in these metrics are significant, with increases of 1.57% and 1.88%, respectively. In vessel structure rehabilitation, VSR-Net excels,

especially in the key metrics of β^{err} and SMD . Compared to other SOTA methods, VSR-Net reduces β^{err} to 7.7409, significantly outperforming methods like DSCNet [27] and GraphConvSeg [36], which have β^{err} values of 8.1602 and 7.6602, respectively. This demonstrates that VSR-Net can more accurately capture and repair vessel structure continuity, avoiding breaking structural integrity. Additionally, in the SMD metric, VSR-Net reduces the error to 0.0343, which is also superior to most SOTA methods, further validating its advantage in maintaining morphological similarity of the vessels. In terms of confidence calibration, VSR-Net further reduces the ECE value to 0.0347, outperforming advanced methods like ATPF [30]. VSR-Net not only provides precise segmentation but also effectively enhances the stability of model predictions, further improving the model's reliability. Finally, in terms of computational efficiency, although VSR-Net is constrained by the complexity of the preceding coarse segmentation models nnUNet and Swin UNETR, it still

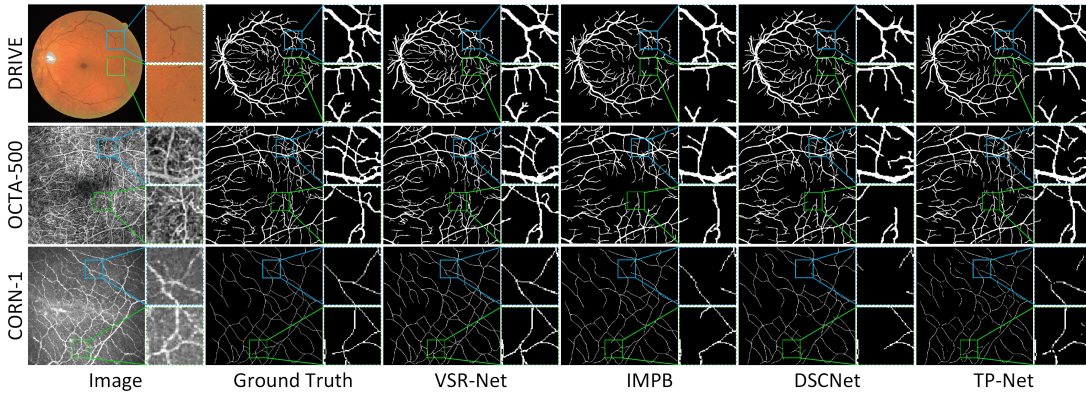


Fig. 7. Visual comparison between vessel-like structure segmentation results by previous methods and vessel-like structure rehabilitation results by VSR-Net in a representative sample from various 2D datasets.

TABLE VI

COMPARISON OF STATE-OF-THE-ART METHODS WITH VSR-Net ON THE ATM [41] TEST SET ACROSS SEGMENTATION, VESSEL REHABILITATION, CONFIDENCE CALIBRATION, AND COMPUTATIONAL EFFICIENCY METRICS

Model	Year	$PA \uparrow$	$Dice \uparrow$	$Jacc \uparrow$	$VBN \downarrow$	$FD \downarrow$	$VT \downarrow$	$\beta^{err} \downarrow$	$SMD \downarrow$	$ECE \downarrow$	$P(M) \downarrow$	$F(G) \downarrow$	$FPS \uparrow$
DeepGraphSeg [14]	2019	0.9183	0.7107	0.5819	0.4556	0.3793	0.5472	10.9209	0.0426	0.0753	29.28	150.34	48
DeepDT [31]	2020	0.9264	0.7233	0.5901	0.2654	0.3351	0.4856	9.5601	0.0399	0.0739	61.71	242.68	42
DARN [59]	2021	0.9262	0.7253	0.5956	0.2606	0.3328	0.4830	11.1203	0.0393	0.0895	43.94	180.39	45
EfficientSegNet [60]	2022	0.9275	0.7254	0.5971	0.2623	0.3121	0.4822	10.6601	0.0388	0.0799	33.28	155.27	49
ERNet [42]	2022	0.9283	0.7293	0.5993	0.2772	0.3016	0.4726	7.9809	0.0383	0.0704	9.91	35.28	58
GraphConvSeg [36]	2022	0.9304	0.7363	0.6045	0.2554	0.2885	0.3706	7.6602	0.0369	0.0852	25.45	135.30	53
CPM [46]	2023	0.9283	0.7309	0.6004	0.2623	0.2988	0.4480	9.6407	0.0380	0.0607	10.88	55.13	50
UR-CarA-Net [61]	2023	0.9293	0.7355	0.6032	0.2606	0.2907	0.3732	9.1604	0.0372	0.0867	22.34	92.64	54
CorSegRec [29]	2023	0.9402	0.7984	0.6682	0.2421	0.2103	0.2645	8.4208	0.0356	0.0604	27.79	180.25	47
DSCNet [27]	2023	0.9418	0.8013	0.6705	0.2387	0.2082	0.2628	8.1602	0.0352	0.0589	30.33	195.71	44
ATPF [30]	2024	0.9435	0.8056	0.6739	0.2354	0.2056	0.2613	7.8408	0.0348	0.0573	32.11	215.32	49
VSR-Net [†]	2024	0.9478	0.8090	0.6799	0.2334	0.1963	0.2607	8.2206	0.0350	0.0569	25.12	115.47	52
VSR-Net [‡]	2024	0.9653	0.8214	0.6927	0.1726	0.1711	0.2509	7.7409	0.0343	0.0347	23.61	138.51	50

* VSR-Net[†] and VSR-Net[‡] are the rehabilitation results based on the coarse segmentation results of nnUNet [62] and Swin UNETR [45], respectively.

TABLE VII

COMPARISON OF STATE-OF-THE-ART METHODS WITH VSR-Net ON THE MIDAS-I [42] TEST SET ACROSS SEGMENTATION, VESSEL REHABILITATION, CONFIDENCE CALIBRATION, AND COMPUTATIONAL EFFICIENCY METRICS

Model	Year	$PA \uparrow$	$Dice \uparrow$	$Jacc \uparrow$	$VBN \downarrow$	$FD \downarrow$	$VT \downarrow$	$\beta^{err} \downarrow$	$SMD \downarrow$	$ECE \downarrow$	$FPS \uparrow$
DeepGraphSeg [14]	2019	0.9354	0.8492	0.7714	0.5123	0.4942	0.6502	9.3402	0.0407	0.0552	66
DeepDT [31]	2020	0.9382	0.8520	0.7735	0.4786	0.4539	0.6204	8.9101	0.0395	0.0534	58
DARN [59]	2021	0.9405	0.8543	0.7759	0.4453	0.4211	0.5907	9.1203	0.0382	0.0498	63
EfficientSegNet [60]	2022	0.9426	0.8567	0.7783	0.4134	0.3928	0.5631	7.9806	0.0373	0.0589	68
ERNet [42]	2022	0.9451	0.8589	0.7810	0.3832	0.3650	0.5378	7.5809	0.0365	0.0517	81
GraphConvSeg [36]	2022	0.9514	0.8650	0.7880	0.3039	0.2902	0.4696	7.6904	0.0433	0.0521	74
CPM [46]	2023	0.9473	0.8610	0.7836	0.3550	0.3384	0.5142	7.2405	0.0387	0.0653	69
UR-CarA-Net [61]	2023	0.9496	0.8635	0.7861	0.3286	0.3134	0.4913	6.9302	0.0417	0.0636	76
CorSegRec [29]	2023	0.9531	0.8672	0.7903	0.2804	0.2681	0.4491	6.5603	0.0338	0.0546	66
DSCNet [27]	2023	0.9547	0.8690	0.7924	0.2587	0.2474	0.4296	6.2308	0.0387	0.0593	61
ATPF [30]	2024	0.9562	0.8708	0.7943	0.2387	0.2280	0.4112	6.9407	0.0437	0.0525	69
VSR-Net [†]	2024	0.9578	0.8725	0.7965	0.2202	0.2105	0.3941	6.0502	0.0327	0.0470	72
VSR-Net [‡]	2024	0.9593	0.8742	0.7982	0.2026	0.1943	0.3778	5.9804	0.0323	0.0459	69

* VSR-Net[†] and VSR-Net[‡] are the rehabilitation results based on the coarse segmentation results of nnUNet [62] and Swin UNETR [45], respectively.

maintains an inference speed of 50 FPS, slightly surpassing newer SOTA methods like CorSegRec [29], DSCNet [27], and ATPF [30]. Its inference efficiency still meets the clinical diagnostic requirements.

Figure 8 presents visual comparisons of 2D vessel-like structure rehabilitation from selected samples of three different 3D vessel-like structures. Two topology-preserving segmentation methods (ATPF [30] and DSCNet [27]) and a deep network-based refinement segmentation method

(GraphConvSeg [36]) are adopted for comparison in terms of maintaining topological integrity and overall segmentation quality. We summarize as follows: (1) VSR-Net demonstrates its superiority in subsection ruptures rehabilitation compared to other topology-based SOTA methods, obtaining more complete end-point segmentation performance. (2) As shown in the second row of Figure 8, VSR-Net and GraphConvSeg [36] both manage to segment certain intracranial arteries that are not clearly marked in the manual

TABLE VIII

COMPARISON OF STATE-OF-THE-ART METHODS WITH VSR-NET ON THE IMAGECAS [43] TEST SET ACROSS SEGMENTATION, VESSEL REHABILITATION, CONFIDENCE CALIBRATION, AND COMPUTATIONAL EFFICIENCY METRICS

Model	Year	$PA \uparrow$	$Dice \uparrow$	$Jacc \uparrow$	$VBN \downarrow$	$FD \downarrow$	$VT \downarrow$	$\beta^{err} \downarrow$	$SMD \downarrow$	$ECE \downarrow$	$FPS \uparrow$
DeepGraphSeg [14]	2019	0.9312	0.8400	0.7582	0.5201	0.4852	0.6403	8.3202	0.0389	0.0534	113
DeepDT [31]	2020	0.9350	0.8443	0.7621	0.4896	0.4503	0.6108	8.0105	0.0375	0.0512	101
DARN [59]	2021	0.9381	0.8472	0.7655	0.4598	0.4204	0.5821	7.5207	0.0364	0.0488	107
EfficientSegNet [60]	2022	0.9405	0.8500	0.7682	0.4309	0.3922	0.5556	7.0103	0.0352	0.0469	114
ERNet [42]	2022	0.9432	0.8530	0.7716	0.4040	0.3659	0.5301	6.7109	0.0342	0.0450	135
GraphConvSeg [36]	2022	0.9484	0.8592	0.7785	0.3331	0.2958	0.4614	5.9505	0.0330	0.0428	124
CPM [46]	2023	0.9450	0.8550	0.7740	0.3787	0.3412	0.5063	6.4507	0.0337	0.0443	119
UR-CarA-Net [61]	2023	0.9468	0.8575	0.7768	0.3550	0.3170	0.4832	6.1903	0.0333	0.0435	129
CorSegRec [29]	2023	0.9500	0.8613	0.7808	0.3125	0.2757	0.4408	5.7204	0.0328	0.0422	110
DSCNet [27]	2023	0.9515	0.8630	0.7829	0.2935	0.2570	0.4212	5.5103	0.0325	0.0417	104
ATPF [30]	2024	0.9530	0.8650	0.7848	0.2757	0.2396	0.4030	5.3107	0.0322	0.0411	115
VSR-Net [†]	2024	0.9542	0.8670	0.7869	0.2602	0.2243	0.3861	5.1506	0.0317	0.0405	123
VSR-Net [‡]	2024	0.9555	0.8690	0.7890	0.2450	0.2101	0.3702	5.0105	0.0313	0.0399	119

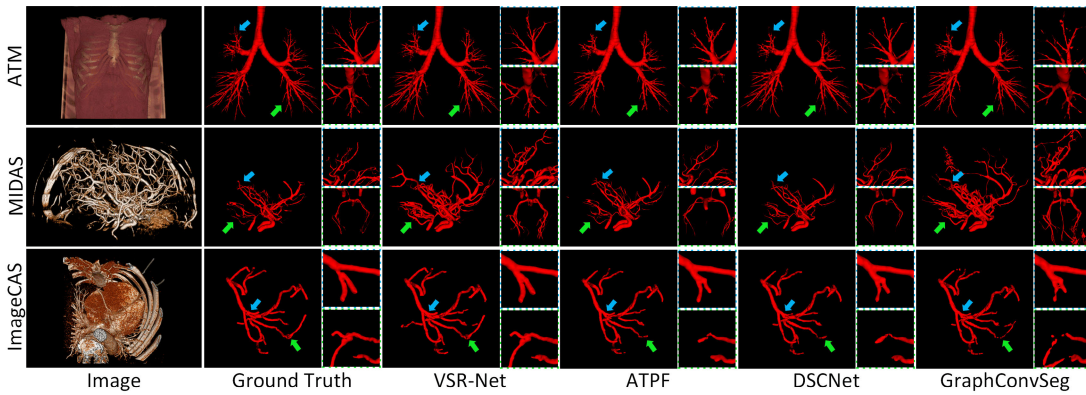
* VSR-Net[†] and VSR-Net[‡] are the rehabilitation results based on the coarse segmentation results of nnUNet [62] and Swin UNETR [45], respectively.

Fig. 8. Visual comparison between vessel-like structure segmentation results by previous methods and vessel-like structure rehabilitation results by VSR-Net in a representative sample from various 3D datasets.

TABLE IX

COMPARISON OF BASELINE METHODS INTEGRATED WITH VSR-NET ON THE DRIVE [7] TEST SET ACROSS SEGMENTATION, VESSEL REHABILITATION, CONFIDENCE CALIBRATION, AND COMPUTATIONAL EFFICIENCY METRICS

Model	$PA \uparrow$	$Dice \uparrow$	$Jacc \uparrow$	$VBN \downarrow$	$FD \downarrow$	$VT \downarrow$	$\beta^{err} \downarrow$	$SMD \downarrow$	$ECE \downarrow$	$FPS \uparrow$
CENet [12]	0.9569	0.8483	0.7184	0.5161	0.6417	0.7660	7.1540	0.0327	0.0603	124
+ VSR-Net ¹	0.9650	0.8864	0.7606	0.1769	0.3834	0.4724	4.0125	0.0157	0.0337	91
+ VSR-Net ²	0.9647	0.8856	0.7595	0.1825	0.3925	0.4815	4.4193	0.0158	0.0338	73
+ VSR-Net ³	0.9645	0.8852	0.7592	0.1870	0.3980	0.4875	4.7284	0.0159	0.0340	58
SKelcon-Net [13]	0.9675	0.8567	0.7205	0.5650	0.6523	0.6509	6.8503	0.0225	0.0592	137
+ VSR-Net ¹	0.9793	0.8916	0.7662	0.1635	0.2237	0.4098	3.6882	0.0149	0.0281	109
+ VSR-Net ²	0.9785	0.8912	0.7660	0.1695	0.2315	0.4175	3.8957	0.0150	0.0283	97
+ VSR-Net ³	0.9783	0.8910	0.7658	0.1748	0.2370	0.4230	4.1024	0.0151	0.0285	85

* + VSR-Net¹, + VSR-Net², + VSR-Net³ represent the results after applying VSR-Net 1, 2, and 3 times, respectively, based on the coarse segmentation results.

annotations, but GraphConvSeg [36] mistakenly segments some other brain tissues as intracranial arteries. (3) Similar to the 2D vessel structure rehabilitation visualizations, the results indicate that VSR-Net excels in 3D vessel-like structure rehabilitation and structure segmentation via graph clustering, proving the generalization and superiority of our method. (4) Unlike 2D structure rehabilitation, 3D structures present the additional challenge of varying extension directions beyond only dense disconnections. The visualization results show that VSR-Net can effectively repair 3D-scale disconnections, accurately determining the extension direction of the break and restoring subsection rupture into a unified structure.

E. Generalization and Reliability Analysis

To further demonstrate the generalization and reliability of our method in 2D and 3D vessel-like structure rehabilitation tasks, we conducted extensive experiments on the DRIVE [7] and ATM [41] datasets. First, we generated coarse vessel-like structure segmentation results using commonly employed 2D and 3D segmentation baselines. For 2D vessel-like structure segmentation, the baselines include CENet [12] and SKelcon-Net [13], while for 3D segmentation, the baselines were nnUNet [62] and Swin UNETR [63]. Based on the coarse segmentation results, we conducted a comprehensive comparison across four dimensions: segmentation, vessel

TABLE X

COMPARISON OF BASELINE METHODS INTEGRATED WITH VSR-Net ON THE ATM [41] TEST SET ACROSS SEGMENTATION, VESSEL REHABILITATION, CONFIDENCE CALIBRATION, AND COMPUTATIONAL EFFICIENCY METRICS

Model	$PA \uparrow$	$Dice \uparrow$	$Jacc \uparrow$	$VBN \downarrow$	$FD \downarrow$	$VT \downarrow$	$\beta^{err} \downarrow$	$SMD \downarrow$	$ECE \downarrow$	$FPS \uparrow$
nnUNet [62]	0.9356	0.7685	0.6150	0.4721	0.3803	0.4817	14.7148	0.0518	0.0820	57
+ VSR-Net ¹	0.9478	0.8090	0.6799	0.2334	0.1963	0.2607	8.2206	0.0350	0.0569	52
+ VSR-Net ²	0.9475	0.8085	0.6794	0.2335	0.1965	0.2609	8.2329	0.0352	0.0571	42
+ VSR-Net ³	0.9473	0.8082	0.6791	0.2337	0.1967	0.2611	8.2454	0.0354	0.0573	36
Swin UNETR [63]	0.9416	0.7772	0.6163	0.4281	0.2939	0.4370	13.8562	0.0486	0.0628	58
+ VSR-Net ¹	0.9653	0.8214	0.6927	0.1726	0.1711	0.2509	7.7409	0.0343	0.0347	50
+ VSR-Net ²	0.9651	0.8209	0.6922	0.1727	0.1712	0.2510	7.7496	0.0345	0.0349	43
+ VSR-Net ³	0.9650	0.8207	0.6920	0.1729	0.1713	0.2512	7.7549	0.0347	0.0351	37

* + VSR-Net¹, + VSR-Net², + VSR-Net³ represent the results after applying VSR-Net 1, 2, and 3 times, respectively, based on the coarse segmentation results.

structure rehabilitation, confidence calibration, and computational efficiency, between the original results and those enhanced with VSR-Net. Additionally, we evaluated the performance after applying VSR-Net multiple times to determine the optimal number of applications for achieving the expected results.

1) *Generalization and Reliability Comparison on 2D Vessel-Like Structure Datasets*: Table IX presents the segmentation, rehabilitation, and ECE metrics for the original 2D vessel-like structure segmentation baselines and their enhanced versions with VSR-Net. The results demonstrate that VSR-Net consistently delivers stable performance improvements across various segmentation baselines. Specifically: (1) From a segmentation perspective, applying VSR-Net once leads to improvements in PA, Dice, and Jaccard indices by 0.0081 to 0.0118, 0.0349 to 0.0381, and 0.0422 to 0.0457, respectively, showcasing the robustness and effectiveness of our method across different baselines. (2) In terms of vessel structure rehabilitation and computational efficiency, VSR-Net significantly improves β^{err} and SMD metrics, reducing β^{err} to as low as 3.6882 and SMD to 0.0149. However, each additional application of VSR-Net results in an FPS reduction range of 12 to 18 frames.

As shown in Figure 9, a single application of VSR-Net can effectively rehabilitate subsection ruptures with minimal risk of excessive rehabilitation. However, Figure 9 also demonstrates that applying VSR-Net multiple times on the coarse segmentation results from SKelcon-Net [13] may cause minor rehabilitation failures. Despite this, it may still manage to rehabilitate subsection ruptures that are unclearly marked in the manual annotations, which requires further validation of their accuracy. Thus, considering both repair effectiveness and computational efficiency, a single application of VSR-Net is typically the most effective strategy, ensuring accurate rehabilitation of subsection ruptures while optimizing computational resources.

2) *Generalization and Reliability Comparison on 3D Vessel-Like Structure Datasets*: Table X showcases the generalization and reliability of VSR-Net in 3D vessel-like structure rehabilitation tasks. Across different segmentation baselines, applying VSR-Net once results in significant performance gains. For example, compared to Swin UNETR [63], VSR-Net improves PA, Dice, and Jaccard indices by 0.0148 to 0.0237, 0.0346 to 0.0442, and 0.0376 to 0.0764, respectively. These

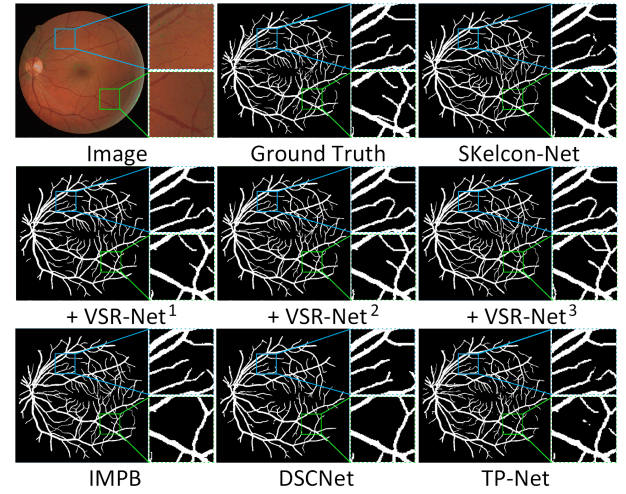


Fig. 9. Visual comparison of vessel-like structure segmentation failures between baselines and VSR-Net in a representative sample from the DRIVE [7] test set. Multiple applications of VSR-Net may lead to over-rehabilitation, including failures to connect subsections that shouldn't be linked. However, a single application of VSR-Net outperforms other SOTA methods in rehabilitating subsection ruptures and preserving topological consistency.

improvements demonstrate the robustness and adaptability of VSR-Net on 3D datasets, enabling more accurate segmentation of complex vessel structures. In terms of vessel structure rehabilitation, VSR-Net also excels, reducing β^{err} to 7.7409 and SMD to 0.0343, highlighting its superiority in maintaining continuity and morphological similarity. However, each additional application of VSR-Net results in an FPS reduction in 6 and 7 frames.

As shown in Figure 10, a single application of VSR-Net can effectively rehabilitate subsection ruptures with minimal risk of excessive rehabilitation. Figure 10 demonstrates that applying VSR-Net multiple times on the coarse segmentation results from Swin UNETR [63] is less likely to cause excessive rehabilitation; However, it may still rehabilitate subsection ruptures unclearly marked in the manual annotations, which requires further validation of their accuracy. Therefore, from the perspective of repair effectiveness and computational efficiency, a single application of VSR-Net is generally the optimal choice, effectively rehabilitating subsection ruptures while maintaining high computational efficiency.

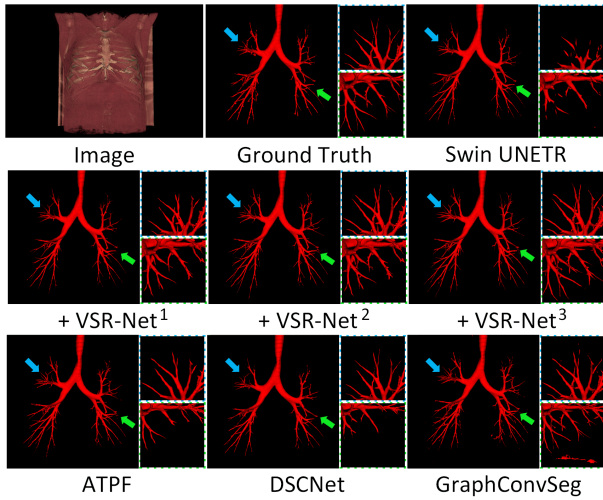


Fig. 10. Visual comparison of vessel-like structure segmentation failures between baselines and VSR-Net, using a representative sample from the ATM [41] test set. These results indicate that while multiple applications of VSR-Net in 3D lung airways might lead to excessive rehabilitation and may segment some unannotated delicate subsections, a single application still outperforms other SOTA methods in terms of accuracy and consistency.

V. CONCLUSION

In this paper, we provide a novel structure rehabilitation perspective to understand and tackle subsection rupture problems in vessel-like structure segmentation tasks by reconsidering it as the vessel-like structure rehabilitation task. Under this view, our proposed VSR-Net can effectively rehabilitate subsection ruptures, preserve the topological integrity, and improve the confidence calibration based on the coarse vessel-like structure segmentation results, by exploring spatial interconnection relationships among subsection ruptures with graph clustering. The extensive experiments on six 2D/3D vessel-like structure datasets manifest the effectiveness of our method.

To further qualitatively and quantitatively demonstrate the generalization and reliability of our VSR-Net, The robust experiments show that our method maintains a well-balanced performance across vessel-like structure segmentation, vessel rehabilitation, confidence calibration, and computational efficiency, consistent with our expectations. However, multiple applications of VSR-Net may lead to over-rehabilitation and produce delicate branch segmentations that were not annotated manually. In future work, we plan to address these challenges and extend the proposed method to unsupervised vessel-like structure rehabilitation tasks, achieving adaptive rehabilitation and providing more theoretical analysis.

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